

Education

M.S.(Research): Electrical Engineering	Indian Institute of Technology (IIT), Delhi — 2025
Bachelor of Technology: Electronics and Communication Engineering	USICT – GGSIPU, Delhi — 2025
Senior Secondary (CBSE): PCM + Computer Science (C++) – 95%	Bal Bhavan Public School, Delhi — 2020

Technical Skills

Languages: C, C++, Java, Python, SQL
Skills: Pandas, NumPy, OpenCV, MATLAB, AI/ML, IOT, VLSI, Antenna Design, Microprocessors (8085/8086) and Controllers, Embedded Systems, Verilog, Communication Systems design, Presentation, Speaking, Management, Leadership.

Research Work

Dynamic Analysis of CCTV Feed via ML Algorithms	2023
Guide: Prof. Ashish Payal, USICT Delhi	Research Thesis Demo
- Implemented adaptive services leveraging facial signal features for real-time predictive state recognition. - Developed a scalable CNN-based classification model with GPU-level parallelism (30% faster, 18% lower memory). - Applied statistical modeling to convert visual cues into time-series datasets for anomaly detection with 92% accuracy. Technologies: Python, OpenCV, PyTorch, Dlib, CNN, ML, PuLP, Statistics, Mathematical Modeling.	

Experience

P3C	2025
Research and Development Engineer. - Designed analog/digital circuits for accurate railway signal detection in real-world environments. - Built signal conditioning and amplification circuits to enhance weak signals and reduce noise. - Deployed and tested solutions on-site, improving detection reliability and system performance.	
Opkey (Product-Based SaaS Company)	2025
AI/ML Backend Developer – R&D - Engineered System Prompt and LLM-driven automation with RAG, increasing reliability by 45% in large-scale AI systems. - Applied SOLID principles and logging in LangChain workflows on Azure that cut tool-call errors by 35% with faster debugging. - Monitored and Optimized Nginx settings that lowered Opkey’s AI tool downtime by 20% while collaborating with UX, frontend and platform teams to gather user requirements and implement backend changes, enabling faster and reliable AI responses. - Automated SAP/Oracle test workflows, cutting manual intervention by 70% and improving test-cycle accuracy by 40%.	
Continental Device India Ltd. (Semiconductor Devices & Chips Manufacturing)	
R&D Engineer - Engineered embedded control and GUI solutions for EV systems, enabling real-time automation and reliable performance. - Programmed and optimized 8/32-bit microcontrollers for low-latency, high-performance operations in constrained environments. - Sole trainee selected to implement predictive maintenance algorithms using sensor data, reducing operational costs by 20%. - Collaborated on scalable, fault-tolerant embedded solutions aligned with industrial standards.	

IBM	2023
Applied Data Scientist - Delivered anomaly detection models that improved forecast accuracy by 30% and processed data ingestion efficiency by 25%. - Analyzed large-scale datasets via Python and SQL that generated actionable insights for 8K+ users. - Optimized data pipelines using lightweight data compression to improve processing and cut storage by 20%.	

Production Projects

AI Personal Assistant “Rockerz” (NLP + CV + Speech Recognition)	Demo — 2025
Guide: Prof. Vandana Nath, USICT Delhi - Built “Assist AI”, a Python assistant with CV, NLP & Brain-Computer Interface for 75% accurate hands-free automation. - Optimize brain signals via filtering, passing through High pass and low pass filters and amplifying with opamp-741 IC to reduce noise.	
Real-Time Object Classifier – OCR Engine (Machine Learning)	Demo 2024
- Trained a custom YOLOv8 model on 1.2K+ labeled images (fruits/vegetables) attaining F1: 0.92 and 97.5% recall. - Boosted real-time processing speed by 20% through targeted model fine-tuning for high-volume industrial use cases. - Achieved 95% OCR accuracy on 500+ samples via adaptive filtering and iterative model refinement.	
Web Gallery Application (Image Gallery)	Live — 2023
- MERN-based image gallery with secure data handling and supporting 1K+ uploads via CRUD for interactive collages. - Enhanced using React, TypeScript, Tailwind CSS, MongoDB, Postman with lazy loading to improve load time by 50%.	

Achievements

- Secured Top 1% Rank (among 100K+) in CodeKaze Contest 2024. Contest Rank Certificate (Top 1%) - HFSS & ORCAD Tools Certification for Antenna Design Issued by Entuple Technologies Pvt. Ltd. - Ranked among Top 6% in GATE ECE 2024 out of 100K+ aspirants nationwide. - Achieved a countrywide Top 5% JEE rank among 1M+ applicants and 1st in School in XII Exams 95%. - Solved 1000+ DSA problems across LeetCode, Code Chef and Specialist at Codeforces. Oracle Certified Professional – Multicloud Architect (OCI–AWS–GCP) and AI Workflow Automation. Certificate	
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